

# Parth Mital

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🐙 github.com/parthmital    🌐 parthmital-portfolio.vercel.app/

## Education & Certifications

### Vellore Institute of Technology (VIT)

B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10)

2023 – 2027

Vellore, India

### OCI 2025 Generative AI Professional

Oracle Certified Professional; ID: 103485063OCI25GAIOCP

July 2026

Valid until July 2028

## Experience

### Concept Marketing

June 2026 – July 2026

Sole Software Developer & Maintainer (Contract)

Vadodara, India

- Migrated legacy ASP.NET ERP to 24/7 Next.js, TypeScript, Prisma, & PostgreSQL on Vercel + Supabase; **ported 114K+ records across 42 tables with zero loss**, replacing a single-PC server.
- Partnered directly with daily staff (6–8 hrs/day) to **replace fragmented tools with unified workflows** for technician dispatch, AMC contracts, billing, inventory variances, & React-PDF invoice exports.
- Secured internet-facing operations with signed cookies, role-based access, Zod validation, & DB rate limits; scheduled **AES-256 encrypted backups to Google Drive** with 30-day retention & cron tasks.
- Engineered test suite with **587 Vitest unit tests & 5 Playwright specs**; delivered live system **managing 2.8K clients, 28K calls, & 60+ users** with ongoing solo production maintenance.

### LearnLine Edustation (IntelGrader)

June 2026 – July 2026

AI Engineer Intern

Remote

- Architected native Android capture app in Kotlin, Compose, CameraX, & OpenCV; implemented **real-time quad stability auto-capture (7 stable frames)**, perspective correction, & web bridge.
- Trained multi-task MobileNetV3 to score image quality & 7 defect flags; applied float16 TFLite quantization, **shrinking model size by 82.5% to 2.19 MB** with **2.55 ms inference latency**.
- Formulated YOLOv8s struck-out text detector during selection; integrated downstream masking to **cut OCR token clutter by 7.11%** while maintaining **80% to 96% semantic similarity**.
- Designed native PyMuPDF parser **extracting 1,033 questions & 383 visuals without GPU**; created dual-GPU EasyOCR cache across **4,574 textbook pages** with AWS Bedrock structuring.
- Drove technical & product decisions with co-founders; deployed **agentic HTML lesson generator across 6 archetypes (axe-core accessibility audited)** & prototype 56s doodle video pipeline.

## Projects & Research

### 🔗 NeuroScan AI: Brain Tumour Benchmark & Clinical Platform

PyTorch, FastAPI, React, VTK.js

- Benchmarked **21 deep learning architectures** across 3 stages in a 2-person team under faculty supervision; co-authored **2 research manuscripts** currently under peer review.
- Trained Stage 1 binary detector (**99.59% accuracy**) & Stage 2 subtyper (**98.58% accuracy**) across 4,705 scans, **eliminating 19.3% cross-dataset leakage** via pHash deduplication.
- Evaluated Stage 3 3D volumetric segmentation on 368 BraTS cases; deployed VNet reaching **0.7655 mean Dice (0.8541 WT)** with only **1.4M parameters & 16.9 min training**.
- Developed full-stack clinical platform for **interactive multi-planar 3D slice navigation**; integrated NiBabel NIFTI parsing, JWT auth, system health checks, & **automated PDF diagnostic reports**.

## Technical Skills

**Languages:** Python, Java, C, C++, SQL, Bash, JavaScript

**AI, ML & Data Science:** Scikit-learn, ONNX Runtime, Hugging Face, Sentence-Transformers, MONAI, timm, Pandas, NumPy

**Frameworks & Web Systems:** Node.js, Tailwind CSS, shadcn/ui, TanStack Query, SQLAlchemy, SQLite, RESTful APIs

**DevOps, Cloud & Tooling:** Docker, Git, GitHub Actions, Linux, CI/CD, Postman, Vite, Agile

**Coursework:** DSA, OS, DBMS, Computer Networks, Software Engineering, Compiler Design, Cloud Architecture, Cryptography & Network Security, NLP, DIP, Linear Algebra, Probability & Statistics